
**A mesocosm experiment assessing the effects of signal crayfish
and artificial light at night on aquatic emergence insects**

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Declaration

I confirm that I am the author of the current work and utilized only the specified tools, explicitly excluding any internet sources not listed in the references. Furthermore, I confirm that I have not previously submitted this work for any other examination procedure. The written version I have presented is identical to the one stored on the electronic storage medium (CD-ROM).

Place: Landau in der Pfalz

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Abstract

Meta-ecosystems are formed when organisms, materials and energy are exchanged across boundaries between ecosystems. Aquatic–terrestrial connections, facilitated by the movement of emergence aquatic insects (EAIs), play an important role in cross-ecosystem exchanges. EAIs, which undergo aquatic larval/pupal and terrestrial flying adult stages, serve as important energy and nutrient sources between aquatic and riverine habitats. They contribute to ecosystem processes such as carbon cycling and nutrient dynamics. However, these important aquatic-terrestrial connections are increasingly threatened by anthropogenic stressors. Two worldwide stressors are the introduction of invasive species such as the signal crayfish (CR) and artificial lighting at night (ALAN). Signal crayfish can disrupt aquatic food webs and degrade habitats, whereas ALAN can interfere crayfish and insect behaviour and community dynamics. Understanding the combined effects of these stressors on EAIs communities is critical for protecting the integrity of ecosystems and the vital services they provide, such as supporting fisheries, contributing to nutrient cycling, and serving as food for terrestrial predators. To study the interactive effects of signal crayfish and ALAN, we conducted a stream mesocosm experiment using 16 separate channels (flume), each 15 meters in length and 1 meter in width. This experiment assessed the effects of these stressors, alone and in combination, on EAIs abundance, biomass, and community composition. Results showed that neither the individual stressors (CR, ALAN) nor their combination (ALAN+CR) had significant effects on the decrease in abundance and biomass among treatments. However, the abundance of EAIs significantly decreased with over time when exposed to either of the stressors alone or in combination. Additionally, the EAIs community composition deviated significantly from the control treatment under the influence of both stressors, whether applied alone or in combination. The observed combined changes of EAIs can be attributed to habitat structure effects by crayfish and behavioural effects induced by ALAN. These findings underscore the importance of considering the interactions between multiple stressors in meta-ecosystems, as they can have complex effects on aquatic insect communities. These impacts can disrupt essential ecosystem functions like nutrient cycling and energy transfer, weakening connections between aquatic and terrestrial habitats. This study highlights the importance of considering the impact of stressors on environmental management, including freshwater habitats and their connections to the terrestrial environment. Further research is needed to understand how different stressors interact in aquatic environments. This includes investigating how aquatic organisms behave and respond to multiple stressors and the long-term effects on ecosystems. Understanding these interactions can help develop better conservation strategies for protecting meta-ecosystems.

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1. Introduction

Ecosystems linked by the spatial movement of materials, energy, and organisms across ecosystem boundaries are referred to as meta-ecosystems (Loreau et al., 2003; Datry, 2021). This idea integrates the views of community, ecosystem, and landscape ecology, going beyond meta-community concepts to include not only organism dispersal but also abiotic constraints and feedback (Polis et al., 1997). Aquatic ecosystems are closely connected to terrestrial ecosystems through the exchange of resources, which shapes species interactions (Nash et al., 2021). Aquatic-terrestrial linkages are mutual exchanges and interactions that occur between aquatic and terrestrial ecosystems (Lamberti et al., 2010). The movement of energy, nutrients and organisms between aquatic and terrestrial ecosystems is crucial. Several examples illustrate this exchange including hydrologic connectivity and cross-ecosystem subsidies contributing to these important exchanges (Bilby 1988; Muehlbauer et al., 2019; Sullivan et al., 2019; Nash et al., 2021). These connections serve as indicators of environmental change and are essential for the exchange of nutrients, energy, and pollutants (Bailey et al., 2004; Williamson et al., 2008; Mota et al., 2014). These linkages involve a network of species interactions, and can be used to identify environmental change and provide management solutions (Sullivan et al., 2019). Emergence aquatic insects (EAIs) such as Diptera (flies), Ephemeroptera (mayflies), Trichoptera (caddisflies) and Plecoptera (stoneflies) play an important role in cross-ecosystem exchanges (Bartrons et al., 2013; Heinrich et al., 2017; Ohler et al., 2022).

EAIs are insects with aquatic larval and/or pupal stages and terrestrial flying adults (Heinrich et al., 2017; Nash et al., 2023). Their development may involve holometabolous metamorphosis (e.g., Diptera, Trichoptera) or hemimetabolous metamorphosis (e.g., Odonata, Ephemeroptera) (Nash et al., 2023). They are found in a variety of freshwater environments, including streams, wetlands, lakes, rivers, and ponds. Several factors can influence their occurrence, including land use, climate, and water quality (Starr et al., 2021). The key orders of EAIs discussed in the literature include Diptera, Ephemeroptera, Odonata, Plecoptera, and Trichoptera, which are known for their significant contributions to ecosystem dynamics and nutrient cycling (Hershey et al., 2010; Shayeghi et al., 2014; Derka et al., 2019). Chironomidae (Diptera) is the most abundant and diverse group found in aquatic ecosystems throughout southern Germany, including both streams and lakes (Salvarina et al., 2017; Ohler et al., 2022). For example, Chironomidae accounted for 87.8% of the insect samples collected in a flood-plain mesocosm (Kolbensschlag et al., 2023). Emergence peaks were determined in the spring, summer, and early autumn seasons in streams in southern Germany (Salvarina et al., 2017; Ohler et al., 2022).

EAIs play an important role in providing vital nutrient support to ecosystems, particularly in the riparian zone. Their contribution is essential to the interconnected food webs within these

ecosystems (Hixson et al., 2015; Twining et al., 2021; Schulz et al., 2023). EAI's play a significant role in aquatic and terrestrial ecosystem processes, such as carbon uptake and nutrient cycling (Recalde et al., 2021). In addition, they play a vital role in energy and nutrient processing, which involves capturing and transporting nutrients from aquatic to terrestrial ecosystems (Gratton & Zanden, 2009; Sute et al., 2015). Moreover, they provide prey and food sources for fish and aquatic predators, as well as amphibians and other terrestrial predators (Hershey et al., 2010). For their physiological well-being, both aquatic and terrestrial animals rely on dietary sources of lipids like polyunsaturated fatty acids (PUFA). These compounds also contribute to secondary production in terrestrial ecosystems (Twining et al., 2016). Because of their high quantities of necessary PUFA, aquatic insects are considered higher -quality as food for terrestrial consumers (Martin et al., 2017; Ohler et al., 2023).

Anthropogenic stressors are environmental pressures and disturbances caused by human activities in ecosystems. Introduction of invasive species, pollution, deforestation, overexploitation of resources, habitat destruction, land use conversion, and pollution are stresses faced by indigenous species (van son et al., 2007; Simeoni et al., 2023). The link between aquatic and terrestrial ecosystems can be disrupted by anthropogenic stressors in several ways, such as those affecting food web responses in the receiving ecosystem (Schulz et al., 2015 and 2023). The impact of these stressors results in modifications to the responses and linkages between the two ecosystems, ultimately causing changes in species composition and distributions (Häder & Barnes et al., 2019). Reduced insect emergence and increased decomposition can be one outcomes of anthropogenic stressors that affect the reciprocal exchanges of materials, energy, and organisms and break strong connections with nearby terrestrial ecosystems (Nash et al., 2021).

One important anthropogenic stressor is invasive species, which are non-native species introduced into new environments, where they can cause substantial harm to ecosystems, the economy, or human health (Leppäkoski et al., 2002; Occhipinti-Ambrogi and Savini 2003; Pimentel et al., 2005; Hyytiäinen et al., 2013). Through a variety of human activities, including boating, fishing, and the release of unwanted aquarium content, invasive species can be introduced both purposefully and inadvertently (Havel et al., 2013). The presence of invasive species can have negative effects on the preservation of native species and the overall ecological balance. These include competing with native species for food and habitat, decreasing the number of predators, and disrupting the functioning of the invaded ecosystem and the food web (Strayer et al., 2010).

Crayfish species are one group of the most prevalent invasive species in freshwater habitats, as they have been introduced in many non-native environments for commercial purposes

(Albertson et al., 2016). Crayfish species are omnivorous in general, with a strong predilection for predatory food. However, when a sufficient supply of macrobenthos is absent, they consume vast amounts of little attractive food, such as macrophytes (Roth et al., 2006). This has substantial adverse effects on ecosystems (Ruokonen 2012; Vaeßen et al., 2015). Additionally, it reduces submerged vegetation and benthic invertebrate populations (Ruokonen 2012; Vaeßen et al., 2015). Significant environmental changes caused by crayfish have negative impacts on ecosystems, including bank collapses, leading to changes in habitat structure, nutrient cycling, and water quality (Piersanti et al., 2018). As keystone species, they frequently induce significant changes across the numerous levels of trophic balance in the ecosystems that they invade (Nunes et al., 2017).

The signal crayfish (*Pacifastacus leniusculus*, Dana 1852) stands out as one of the most extensively distributed invasive crayfish species (Galib et al., 2021; Sanders et al., 2022). Signal crayfish, originally from North America, were identified as an invasive species in the UK in the 1970s (Freeman et al., 2010; Larson and Olden 2011; Ruokonen 2012; Vaeßen et al., 2015). The introduction of the signal crayfish led to the elimination of the indigenous white-clawed crayfish species and a reduction in the numbers of young salmon fish being recruited into local river systems, as documented in studies by Peay et al. (2009) and Freeman et al. (2010). Signal crayfish are primarily nocturnal species that hide during the day and become more active at night as a result of this behaviour (Johnson et al., 2014; Thomas et al., 2016). Similar nocturnal behaviours could be an adaptation to avoid predators (Grober et al., 1998). As a widely distributed species found in many lakes and rivers, signal crayfish have a major impact on ecosystem nutrient cycles, biodiversity, sedimentation, and water quality (Yu et al., 2018). They can transmit new diseases to native species by acting as parasite vectors (Crowl et al., 2008). Riverbeds can modify in many ways by signal crayfish, and their sediment transport is accelerated. Signal crayfish can link to changes in native macroinvertebrate communities, including decreases in abundance and richness (Usio et al., 2009; Bjurström et al., 2010; Ruokonen 2012; Mathers et al., 2020; Galib et al., 2021; Watanabe et al., 2022). Signal crayfish in English rivers have led to significant changes in benthic macroinvertebrate ecosystems (Mathers et al., 2016). Long-term studies found that signal crayfish had broader ecological impacts than short-term studies, as long-term observed a decrease in the abundance of more taxonomic groups (McCarthy et al., 2006).

Besides the occurrence of invasive species, light pollution represents another widespread common stressor in freshwater ecosystems (Hölker et al., 2023). Light pollution can result in disturbances to natural light conditions, impacting the regulation of biological processes in aquatic organisms and potentially disrupting the stability of ecosystems (Hussein et al., 2021;

Mayer-Pinto et al., 2022). Artificial lighting at night (ALAN) refers to the illumination produced by artificial light sources during nighttime hours and is included in environmental light pollution (Gaston et al., 2015; Sullivan et al., 2019). Many human population centres are located close to water bodies, making aquatic ecosystems and their associated environments, such as riparian zones, susceptible to the effects of ALAN (Parkinson et al., 2020). In recent years, there has been an increase in studies about the global increase in ALAN and its effects on ecosystems and organisms (Hölker et al., 2010; Manfrin et al., 2017a & 2017b). ALAN can limit the migration of aquatic invertebrates into terrestrial systems, disrupt insect navigation and pollination, and making insects particularly vulnerable (Warrant et al., 2011; Knop et al., 2017; Firebaugh et al., 2019; Larsson et al., 2020; Parkinson et al., 2020). Additionally, ALAN can lead to a rise in the number of terrestrial invertebrate predators that rely on aquatic food to supplement their diet (Manfrin et al., 2017). ALAN has a significant impact on both aquatic and terrestrial organisms, both individually and at population levels (Meyer et al., 2013). Therefore, energy transfer from aquatic to terrestrial habitats and trophic dynamics may be influenced by ALAN (Neil and Wu, 2006; Bennie et al., 2018; Sullivan et al., 2019). According to research, ALAN significantly affects insect communities, resulting in reduced abundance, richness and changes in community composition (Meyer et al., 2013; Muehlbauer et al., 2014; Perkin et al., 2014a; Perkin et al., 2014b; Manfrin et al., 2017; Owens et al., 2020; Grubisic et al., 2021; Kehoe et al., 2022). ALAN has led to significant changes in benthic macroinvertebrate ecosystems in Texas rivers, resulting in a notable decline of 37% in key taxa, including a 58% decline in Simuliidae and a 51% decline in Baetidae (Henn et al., 2014). Ecological effects of ALAN may not be apparent in the short-term but long-term effects on insect communities may be visible (van Grunsven et al., 2020).

In the past decade, there has been a rise in interest in studies examining how multiple stressors affect aquatic ecosystems through meta-analyses and experimental methods (Feckler et al., 2018; Bray et al., 2019; Bundschuh et al., 2020; Schulz et al., 2023). Multiple stressors refer to interactions in which the combined effects of two or more stressors are similar (additive), greater (synergistic), or less (antagonistic) than expected based on their individual effects (Folt et al., 1999; Jackson et al., 2016; Orr et al., 2020). A recent study describes that combined stressor effects were primarily additive suggesting that knowledge of single-stressor effects in these stream communities can be used to predict responses to multiple stressors (Elbrecht et al., 2016). Understanding the combined effects of multiple stressors is important because they have the potential to cause ecological disturbances in aquatic ecosystems (Ormerod et al., 2010; Piggott et al., 2012).

Multiple studies have shown that both signal crayfish (Galib et al., 2021) and ALAN (Grubisic et al., 2021) reduce insect abundance independently. However, Thomas et al. (2016) determined that ALAN reduces the nocturnal activity and interspecific interactions of signal crayfish. Therefore, the effect of signal crayfish may be reduced in the presence of ALAN, leading to an antagonistic effect on aquatic insect communities.

In this thesis, data were collected continuously for six weeks (from May 2022 to July 2022) during the initial emergence period to determine differences in abundance, biomass, and composition of aquatic insects in the mesocosm experiment. In addition, this thesis aims to examine how the presence of signal crayfish, ALAN exposure, and their interaction affected the abundance, biomass, and composition of emergence insect communities.

There we expected for EAI communities: (i) EAI abundance and biomass decrease with respect to the control in the presence of signal crayfish, followed by ALAN and then the both stressors combined. (ii) abundance decrease is more pronounced after the long-term (week 6) than the short-term (week 1) in the presence of signal crayfish, followed by ALAN and then the combined stressors. (iii) emergence insect composition deviated most from control after the long-term (week 6) than the short-term (week 1) in the presence of signal crayfish, followed by ALAN and then combined (ALAN+CR) stressor.

2. Methodology

2.1 Experimental setup

The Riparian Stream Mesocosm (RSM) is an outdoor facility. The RSM is situated adjacent to the river Queich in Rhineland-Palatinate, Germany (49°12.067'N/8°8.338'E).

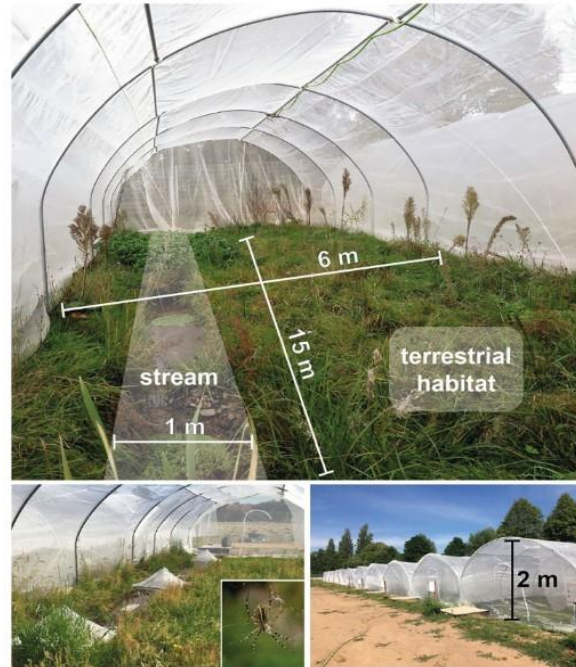


Figure 1: A riparian stream mesocosm (RSM) with an artificial flume and surrounding terrestrial habitat was comprised of 16 replicated mesocosm units (15 m by 6 m) in Landau, southwest Germany. (Taken from Manfrin et al. 2023)

RSM is composed of 16 separate units, each of which consists of an artificial flume measuring approximately 15 m in length and 1 m in width, and a nearby terrestrial area measuring approximately 15 m in length and 4 m in width. To prevent flying insects from transferring between units, each RSM unit was outfitted with a frame structure, with a maximum height of 2 m, covered in a 1 mm mesh gauze. Water from the Queich river is used to supply each flume separately in flow-through. To assess the effect of ALAN on aquatic communities, 8 mm thick 4000–4500 K neutral-white 24V COB LED strips (LED Universum, Dresden, Germany) were employed, powered by a dimmable 320 W LED-Trafo (HLG320H-24B, Min Well, Amterland). The LED strips were set inside cable channels fixed 1m above the water surface. Light was set from approximately 9 p.m. to 6 a.m. and light intensity at 20 ± 2 lux. Timings are checked twice a week and adjusted if necessary. To evaluate the impact of invasive signal crayfish presence, crayfish were collected daily from April to mid-May from the Queech River using crayfish traps with funnel entrances as cat food baits (Peay et al. 2009). Traps were set out in the evening and taken out in the early hours of the morning. Crayfish that had been collected were kept alive

until the experiment began in a 5000 L tank filled with constantly flowing water from the Queich River. Twenty-seven crayfish were deployed at the beginning of the experiment, with equal distributions of size and sex. In addition, regardless of treatment, nine 15 cm PVC tubes were put in each flume and secured with wire ties to serve as crayfish shelters. Every week, in the evening presence of crayfish was checked with flashlights in all of the flumes, and any dead ones were eliminated from the system. At the end of the experiment, all crayfish were removed from the system and final density approximately 0.7 ind/m². The experimental design included four replicates per treatment control (no stressors), crayfish, ALAN, and the combination of ALAN + crayfish. Within this setup, Flumes 4, 7, 10, and 13 are Control, Flumes 2, 5, 12, and 15 are Crayfish, with varying crayfish numbers (7, 9, 10, 11), Flumes 3, 6, 9, and 16 are ALAN, Flumes 1, 8, 11, and 14 are ALAN + CR with varying crayfish numbers (15, 10, 7, 11). This randomized-block design allowed for a nuanced investigation into the effects of different stressor combinations, encompassing CO, ALAN, CR, and ALAN + CR across the entirety of the experiment. The stressor phase of the experiment was set by the deployment of signal crayfish and the initiation of light at night on the 24th of May 2022. The end of the stressor phase was finalised by the removal of all individuals of signal crayfish on the 6th of July 2022, with a total duration of six weeks of exposure to the stressors.

2.2 Emergence insect collection and identification

There was a slight variation in the sampling flumes width, measuring 0.7 m² for the upstream, 0.75 m² for the midstream, and 0.8 m² for the downstream trap instead of 1 m width. As a result, less than one square meter was covered by each emergence trap (Figure 2). Three floating pyramidal emergence traps (Cadmus et al., 2016) were installed in each RSM flumes. The emergence traps were positioned upstream, midstream, and downstream of each artificial flume to collect the emergence insect (Figure 2). The traps were emptied twice a week during the whole duration of the experiment. In addition, a pre-sampling was performed, one week before the initiation of the stressors / stressor phase to assess the initial abundance of emerging insects. Samples were frozen immediately after sampling and stored at – 20°C until processing and analysis. Collected frozen emergence insects were identified to family level using stereomicroscope magnification. The emergence insects were identified using the following reference books: Nilsson (1996a, 1996b) and Merritt and Cummins (1996). For Chironomidae, individuals were separated into Tanypodinae and non-tanypodinae specimens, as their larvae show different feeding strategies (Tachet et al. 2010). Identified emergence insects were dried using a freeze-dryer for 48 hours and weighed to the nearest 0.01 mg. There were occasional differences in the number of collection days between samples. As a result, the biomass and

abundance of each sample were standardised for each trap area and total number of collection days.

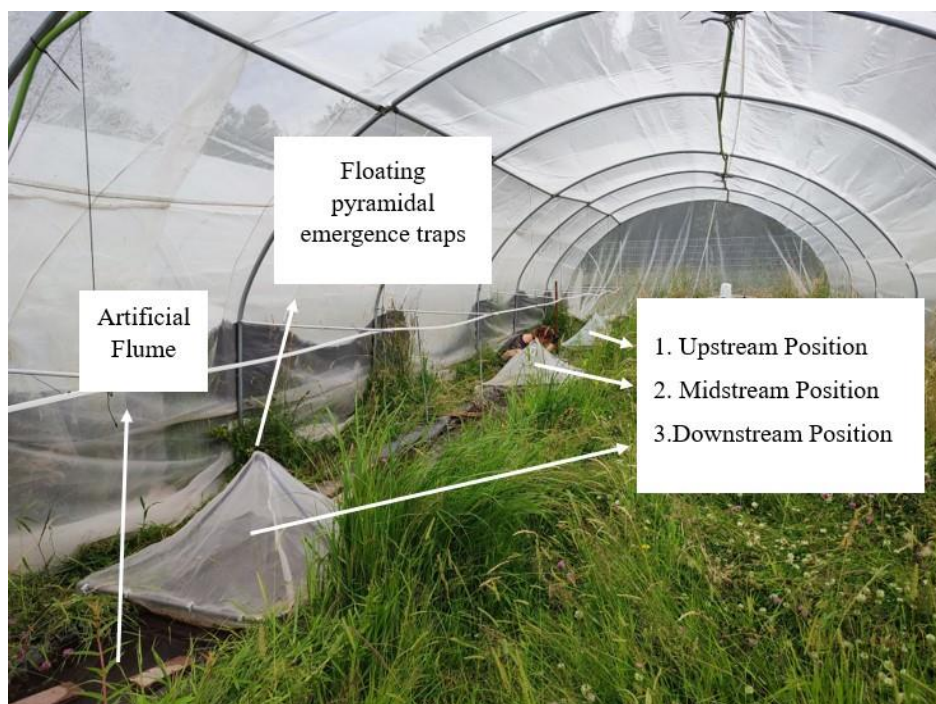


Figure 2: General design inside of artificial flume in RSM

2.3 Data analysis

The experiment examined the emergence rate of emergence insects abundance and biomass over six weeks (week_1 to week_6) during stressors were applied. For this experiment, sampling was conducted at four key time points.: Week_0: A pre-sampling to determine baseline emergence rates (ind/m²/d) before the stressors are introduced. Biomass data for the pre-sampling was not available. Week_1 sample after one week of stressor exposure to assess the short-term response, Week_4 mid-term and Week_6 to evaluate the long-term response. To calculate the daily abundance emergence rate (ind/m²/d) and the daily biomass emergence rate (mg/ m²/d) is expressed.

To investigate the influence of stressors on emergence insects, a linear mixed-effects model (LMMs) with a Gaussian (normal) distribution was used. The R-package lme4 (version 1.1.35.1, Green 2009) was used to fit LMMs. The models employed a Gaussian (normal) distribution and an identity link function. Fixed effects included (treatment, week, and their interaction), and flume was included as a random effect. The response variables abundance and biomass rate were log transformed for these analyses to meet the homoscedasticity and normality assumptions. Furthermore, the pseudo-R-squared value for LMMs was calculated to estimate the proportion of variance explained by the fixed and random effects. These diagnostic checks collectively contribute to a thorough evaluation of the model's assumptions and overall

performance in capturing the underlying patterns in the abundance and biomass data. Furthermore, the analysis includes the determination of the linear mixed-effects model's overall significance through an analysis of variance (ANOVA). To delve deeper into the differences between levels of factors, least-square means (lsmeans) were computed for the interaction between 'Week' and 'Treatment' using the 'lsmeans (2.30.0 Lenth,2018)' package in R. Pairwise comparisons were performed between specific levels of factors, and a detailed summary with inferential statistics was obtained.

To further examine changes in community composition of abundance, Bray-Curtis dissimilarities with permutational multivariate analysis of variance (PERMANOVA) in 999 permutations were used. Logarithmic transformation (Clarke and Warwick, 2001) of the abundance data is applied to compress the scale, giving more weight to less abundant species and helping to normalize skewed abundance data. PERMANOVA analysis was performed to investigate the overall impact of Treatment, Week along with their interactions, on community composition. The results provide insights into the degree of dissimilarity and dispersion in the community structure. PERMANOVA results provide information on the explained variance and the significance of each factor of abundance dataset. To further investigate the interaction between Treatment and Week pairwise PERMANOVA comparisons are conducted using the 'pairwiseAdonis (0.4.1, Martinez Arbizu 2020)' package in R. Non-metric Multidimensional Scaling (NMDS) is employed to analyse and visualize changes in community composition. The NMDS results are presented using the 'ggplot2 (3.4.2, Wickham 2016)' package, with points differentiated by Treatment, shaped and coloured accordingly.

All calculations, statistics and data visualizations were conducted with R (4.3.3, R Core Team, 2024). Additionally, statistical analysis methods and paraphrasing were developed using ChatGPT (OpenAI, 2024).

3. Results

From May to July, a total of 62,112 aquatic emergence insects, representing 23 families, were collected. The total biomass of each taxon is expressed as milligrams (mg). Chironomidae (Diptera) was the most prevalent family showing an abundance of 60,770 individuals, representing 97.84% of the collected specimens. In terms of biomass, Chironomidae has a total biomass of 12478.79 mg, constituting 93.67% of the total biomass in the samples (Table S1).

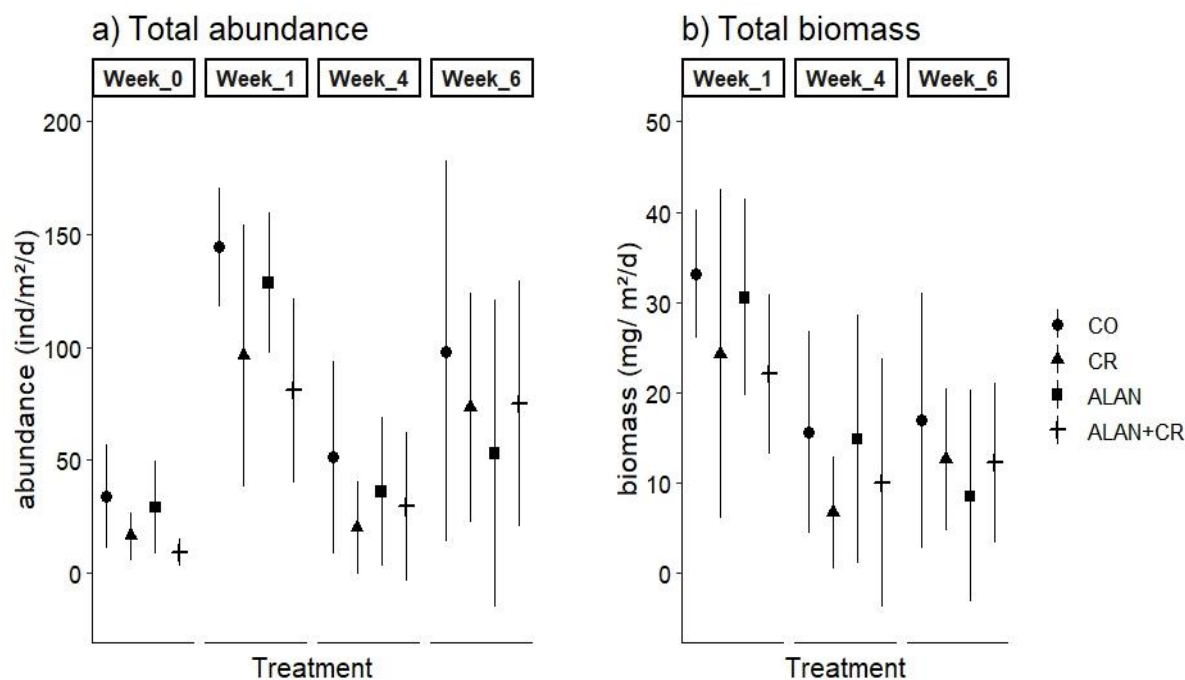


Figure 3: a. abundance rate (ind/m²/day) b. biomass rate (mg/m²/day) across treatments.

Mean values are displayed with different symbols: circles =CO; triangles =CR; squares =ALAN; plus sign =(ALAN+CR).

3.1 Effects of stressors on the emergence insect's abundance

Analysis of variance (ANOVA) explored the main effects of signal crayfish, ALAN, and their interaction on the EAI's abundance. Treatment was not found to be significant ($F = 2.74$, $p = 0.106$), indicating that experimental stressors did not impact EAI's abundance (Table 1).

Table 1: ANOVA Table based on the linear mixed-effect model for the abundance of EAI's exposed to stressors. P-values <0.05 are printed in bold. Where, degrees of freedom (df), sum of squares (SS), mean square (MS), F- values, and p-values were calculated

Factor	df	SS	MS	F value	p value
Treatment	3	1.84	0.614	2.74	0.106
Week	3	29.1	9.67	43.1	<0.0001***
Treatment:Week	9	3.42	0.380	1.69	0.222

Based on this ANOVA result, the picture becomes less clear when examining the p-values ($p > 0.05$), no significant differences were detected between the treatment groups (Table 1, Figure 3a). Data did not align with hypothesis 1, stressor impacts on EAI's abundance.

The factor week showed a significant effect on EAI's abundance ($F = 43.1$, $p < 0.0001^{***}$). This temporal influence suggests dynamic changes in the EAI's abundance over the experimental period (Table 1). Pairwise comparison will give a details clear idea about significant impact (Table 2).

Table 2: Pairwise comparison for treatment, week, and their interaction on abundance, P-values < 0.05 are printed in bold. Where, estimated parameters, standard error (SE) degrees of freedom (df), t- statistic, and p-values were calculated.

Treatment	Contrast	Estimate	SE	df	t,ratio	p,value
CO	Week_0 - Week_1	-1.49	0.335	36	-4.46	0.0004
CO	Week_0 - Week_4	-0.314	0.335	36	-0.937	0.785
CO	Week_0 - Week_6	-0.935	0.335	36	-2.79	0.039
CR	Week_0 - Week_1	-1.72	0.335	36	-5.12	0.0001
CR	Week_0 - Week_4	-0.072	0.335	36	-0.215	0.996
CR	Week_0 - Week_6	-1.43	0.335	36	-4.27	0.0008
ALAN	Week_0 - Week_1	-1.52	0.335	36	-4.55	0.0003
ALAN	Week_0 - Week_4	-0.163	0.335	36	-0.488	0.961
ALAN	Week_0 - Week_6	-0.445	0.335	36	-1.33	0.551
ALAN+CR	Week_0 - Week_1	-2.12	0.335	36	-6.32	<.0001
ALAN+CR	Week_0 - Week_4	-0.956	0.335	36	-2.85	0.034
ALAN+CR	Week_0 - Week_6	-2.00	0.335	36	-5.98	<.0001

The investigation into the abundance dynamics of EAI's under different time (weeks) intervals revealed notable patterns. Significant differences in EAI's communities were observed in the presence of signal crayfish from week 0 to week 1 ($p\text{-value} = 0.0001$), week 6 ($p\text{-value} = 0.0008$), and not at week 4 ($p\text{-value} = 0.996$). The abundance of EAI's showed a significant increase when exposed to ALAN from week 0 to week 1 ($p = 0.0003$), but there were no significant differences at week 4 and 6. The combination of ALAN and crayfish showed significant differences at all time points (week 1: $p\text{-value} < 0.0001$, week 4: $p\text{-value} = 0.034$, week 6: $p\text{-value} < 0.0001$), showing the less increase in emergence in week 1 compared to stressors alone (Table 2 & S3, Figure 3a). In summary, although the observed data generally aligned with the hypothesis 2 regarding pronounced reductions in EAI's abundance in the long-term presence of crayfish and combined stressors, there was disagreement with the hypothesised effects of ALAN.

PERMANOVA analysis, explored the main effects treatment was determined to be significant ($F = 2.24$, $p = 0.027$), indicating that experimental stressors had a discernible impact on EAI community composition. The factor Week showed a significant effect on EAI community composition ($F = 17.1$, $p < 0.001$). This temporal influence suggests dynamic changes in the EAI communities over the experimental period. The interaction between Treatment and Week was not found to be significant ($F = 1.38$, $p = 0.130$). Treatment explained 6% and week explained 43% of the observed dissimilarities between groups (Table 3).

Table 3: Treatment and week effects on the community composition using PERMANOVA showing degrees of freedom (Df), sums of squares (SS), R-squared, F value, p-values. P-values < 0.05 are printed in bold. Number of permutations = 999

Factor	Df	SS	R ²	F	Pr(>F)
Treatment	3	0.141	0.057	2.24	0.027*
Week	3	1.07	0.433	17.1	0.001 ***
Treatment: Week	9	0.259	0.105	1.38	0.130
Residual	48	1.01	0.405		
Total	63	2.47	1.00		

Graphically, plot visually represents the separation of groups (CO, ALAN, CR, ALAN + CR) based on community composition. Each group is distinguishable, suggesting distinct effects on EAI communities. Stressors was represented by a unique combination of shape and colour, facilitating the identification of treatment-specific clusters (Figure 4).

Pairwise PERMANOVA Comparisons will give a details clear idea about significant impact. Notably, crayfish separated from week 0 control state over time, with week 6 communities deviating the most from control at week 6 ($p\text{-value} = 0.036$) and week ($P\text{ value} = 0.033$).

Table 4: Pairwise PERMANOVA Comparisons for treatment, week, and their interaction on community composition; sums of squares (SS), R-squared, F value, p-values. P-values < 0.05 are printed in bold.

Treatment	Pairs	SS	F. Model	R ²	p.value
CO	Week_0 - Week_1	0.114	7.31	0.549	0.028
CO	Week_0 - Week_4	0.09	3.61	0.375	0.079
CO	Week_0 - Week_6	0.083	3.78	0.386	0.079
CR	Week_0 - Week_1	0.158	11.9	0.666	0.035
CR	Week_0 - Week_4	0.056	1.97	0.247	0.187
CR	Week_0 - Week_6	0.177	12.5	0.675	0.036
ALAN	Week_0 - Week_1	0.151	12.5	0.676	0.022
ALAN	Week_0 - Week_4	0.089	4.49	0.428	0.034
ALAN	Week_0 - Week_6	0.039	1.96	0.246	0.203
ALAN+CR	Week_0 - Week_1	0.293	13.7	0.696	0.035
ALAN+CR	Week_0 - Week_4	0.151	4.029	0.401	0.054
ALAN+CR	Week_0 - Week_6	0.309	11.3	0.654	0.032

Combined stressor followed a similar pattern to CR, with week 6 communities being more dispersed and different from control at week 6 (p-value = 0.032) and week 1 (p-value = 0.035). ALAN suggest some dispersion at weeks 1 and 4, indicating deviation from the control over time. However, there was no significant difference between ALAN weeks 6 (p = 0.203), which is contrary to the hypothesis of a greater deviation after the long term (Table 4 & S5).

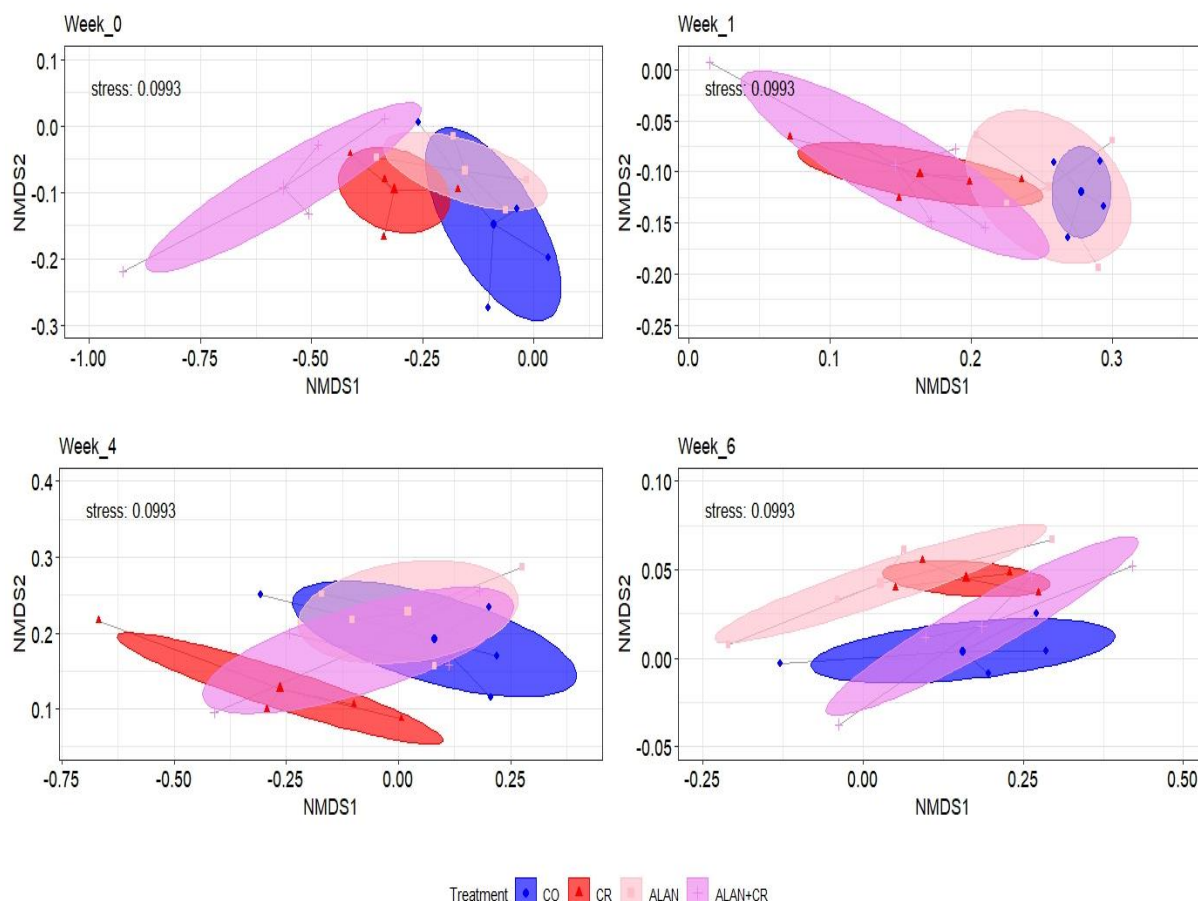


Figure 4: NMDS plot showing the clustering of EAI communities under different treatments over time. Each point represents a sample, coloured and shaped according to the treatment. Confidence ellipses illustrate the dispersion of samples within each treatment group. Each facet represents a specific time point, providing insights into the dynamic response of EAI communities to stressors over time. circles =CO; triangles =CR; squares =ALAN; plus sign =(ALAN+CR).

In summary, the NMDS plot and pairwise PERMANOVA results from Table 3 & 4 together support the hypothesis that invasive crayfish (CR) deviated the most from the control after long-term treatment (week 6). The combined stressor (ALAN + CR) also showed substantial deviation, the same extent as the CR at week 6. ALAN alone did not deviate as much as expected after long-term follow-up.

3. 2 Effects of stressors on the emergence insect's biomass

Analysis of variance (ANOVA) explored the main effects of signal crayfish, ALAN, and their interaction on the EAI's biomass. Treatment was not found to be significant ($F = 0.913$, $p = 0.489$), indicating that experimental stressors did not impact EAI's biomass. Based on this ANOVA result, the picture becomes less clear when examining the p-values ($p > 0.05$), no significant differences were detected between the treatment groups (Table 5, Figure 3b). Data did not align with hypothesis 1, stressor impacts on EAI's biomass.

Table 5: ANOVA Table based on the linear mixed-effect model for the biomass of EAI's exposed to stressors. P-values < 0.05 are printed in bold. Where, degrees of freedom (df), sum of squares (SS), mean square (MS), F- values, and p-values were calculated

Factor	df	Sum Sq	Mean Sq	F value	p value
Treatment	3	0.491	0.163	0.913	0.489
Week	2	8.76	4.38	24.4	0.001***
Treatment:Week	6	1.84	0.308	1.71	0.265

4 Discussion

The present study investigated the combined effects of two widespread anthropogenic stressors on EAI: the signal crayfish and ALAN. The hypothesis suggested that EAI abundance and biomass would decrease the most in the presence of crayfish, followed by ALAN exposure, and then the combined stressor treatment. Furthermore, abundance reductions and community composition changes were expected to be more pronounced after long-term (week 6) exposure compared to short-term (week 1) exposure, particularly for CR followed by ALAN and then ALAN + CR. By investigating these hypotheses, this thesis aimed to shed light on the potential interactions between these stressors and their cumulative impact on EAI communities.

4.1 Stressors effect among treatment

The experiment did not support for hypothesis 1, revealing no effects of signal crayfish, ALAN exposure, and their interactions on EAI communities. EAI abundance and biomass show trends of decrease, particularly the abundance of EAI exposed to crayfish during week 4, however not significant (table 1 & 5). Although several studies have suggested that the presence of signal crayfish negatively affects aquatic insects, these findings are inconsistent with those of Crawford et al. (2006), who found no decrease in the overall biomass of invertebrates in regions with signal crayfish on the Clyde River in Scotland.

For signal crayfish, their effects on EAI abundance and biomass can be modulated by environmental conditions. For example, temperature and flow affect the movement and dispersal of signal crayfish in rivers, indicating that environmental factors may limit their spread and consequent effect on aquatic insects (Bubb et al., 2004). Signal crayfish exhibit a diverse diet consisting of macroinvertebrates, periphyton, and detritus, with seasonal variation. This dietary change enabled them to coexist with aquatic insects without directly affecting them (Ercoli et al., 2021). The density of signal crayfish populations varies how they affect aquatic insects. They have little effect on insect populations in low-density areas (Hudina et al., 2022). Regarding ALAN, the lack of significant impacts on EAI abundance and biomass could be attributed to several factors like to the intensity, duration, and characteristics of the light source. For example, Insects and other invertebrates may be less attracted to ALAN in freshwater environments when applied with low intensity and duration (Grubisic 2018; Ganguly & Candolin 2023). In addition, vegetated buffer zone might have reduced the incidence of ALAN since this zone absorbs and blocks some light (Ganguly & Candolin 2023), growth of riverine vegetations was recorded during present study.

Contrary to expectations, the combined stressor (ALAN + CR) did not consistently exhibit the strongest negative effects on EAI abundance and biomass. This finding, suggests a possible

lack of synergistic or additive effects between these two stressors on EAI abundance and biomass. The observed inconsistencies in the effects of stressors on EAI abundance and biomass highlight the need for further exploration.

4.2 Stressors effect at different time points

The present study determined significant changes in EAI over time, showing temporal dynamics of EAI in stream. Particularly, it was determined a significant increase in the abundance of EAI from week 0 to week 1 in all treatments, indicating a natural pattern of emergence in coincidence with the spring peak. Both CR and combined stressors at 6 weeks showed significant decrease in long-term effects compared to controls, primarily supporting hypothesis 2 of the present study. Regarding ALAN exposure, a short-term effect was observed at week 1 compared to controls; This effect was not maintained in subsequent (weeks 4 and 6); Therefore, the long-term effect could not be distinguished, which deviates from the primary hypothesis 2 of the present study (Table 2). Long-term effects of ALAN could not be distinguished, possibly because the vegetated buffer zone may reduce the incidence of ALAN as the zone absorbs and blocks some light (Ganguly & Candolin 2023), the growth of riparian vegetation was recorded in the present study.

The present results are in line with Ohler et al. (2022) showing the effects of land use on emergence. Different aquatic insect species emerge at different times and follow different seasonal patterns, which is important for population dynamics in interconnected terrestrial systems (Nakano & Murakami, 2001; Ohler et al., 2022). Population biomass, maturation rate, and predator growth rate are all affected by emergence time (Ohler et al., 2022). Temporal changes in the abundance of EAI in the presence of signal crayfish and ALAN can be attributed to several factors.

The predation pressure exerted by signal crayfish plays a significant role, leading to a decline in the abundance of macroinvertebrate taxa over time, thus affecting EAI abundance (Guan and Wiles, 1998; Aroviita and Hämäläinen, 2008; Elliot, 2008; Ruokonen et al., 2016). Moreover, the presence of crayfish interacts with environmental variables like water temperature and level to influence temporal variations in macroinvertebrate abundance (Bjurström et al., 2010; Ruokonen et al., 2016). Similarly, insects and other organisms exhibit negative physiological effects from ALAN, which are mediated by temporal changes in the composition of mature assemblages in aquatic environments (Marangoni et al., 2022). Additionally, ALAN can affect the behaviour of aquatic invertebrates over time, such as their movements, habitat choice, and foraging activities. Thus, affecting their abundance and the dynamics of their interactions with other species (Ganguly & Candolin 2023). Moreover,

ALAN can alter the feeding rates, absorption efficiency, and growth rates of EAIs, further affecting their abundance over time (Marangoni et al., 2022).

Overall, experimental evidence supporting the hypothesis 2 that the combined presence of signal crayfish and ALAN significantly effect the abundance of EAIs, with effects that are more pronounced over time and when both stressors are present simultaneously. However, this reduction was most pronounced in the ALAN+CR group. This suggests that the two stressors interact synergistically, producing greater declines than expected from their individual impacts. According to Jackson et al. (2016), synergistic interactions between multiple anthropogenic stressors are common in aquatic ecosystems. These results underscore the importance of considering multiple stressors in ecological studies, and highlight the need for further research to understand the mechanisms driving the observed interactions between invasive species and anthropogenic stressors in meta-ecosystems.

4.3 Stressors effect in community composition

This study investigated the combined effects of signal crayfish and ALAN on EAIs community composition. Results supported the overall hypothesis 3, demonstrating that the stressors group had a significant effect on the EAIs community compared to control. Interestingly, the time factor also played an important role, with EAIs community composition exhibiting dynamic changes over the studied period. The presence of signal crayfish exhibited the most pronounced deviation from control, especially at week six. This suggests a potentially strong and time-dependent effect of crayfish presence on EAIs communities. The combined stressors treatment (ALAN+CR) followed a similar pattern, although the deviation from control was less pronounced than the individual crayfish treatments. In contrast, communities exposed only to ALAN showed some scatter in the weeks 1 and 4, indicating deviation from control conditions over time. However, contrary to the initial hypothesis, there were no significant differences between the ALAN community and control at week 6 (table 4).

The deviation in EAIs community composition in the presence of signal crayfish might be related to the selective predation of signal crayfish, leading to changes in the abundance and distribution of specific taxa (Usio et al., 2006; Lagrue et al., 2014; Ruokonen et al., 2016; Mathers et al., 2020). Additionally, signal crayfish alter their habitat through burrowing activity, which alters the physical properties of the riverbed. These habitat changes directly affect the insect community (Rice et al., 2014; Faller et al., 2016; Mathers et al., 2020). Furthermore, the presence of signal crayfish can disrupt trophic dynamics within insect communities, affecting predator-prey relationships. This disruption can cascade within the community, leading to community change (Usio et al., 2006; Ruokonen et al., 2016; Mathers et al., 2020). Competition

for resources is another significant factor, as signal crayfish compete with insects for food and habitat resources, affecting insect abundance and distribution (Usio et al., 2006; Ruokonen et al., 2016). Finally, indirect effects such as changes in water quality and habitat complexity associated with signal crayfish presence may influence insect community composition (Usio et al., 2006; Lagrue et al., 2014; Ruokonen et al., 2016; Mathers et al., 2020).

Similarly, in the presence of ALAN might be related to the changes in community composition caused by taxon-specific reactions to ALAN drive some species to benefit from increased light as a resource for detecting prey, while others experience decreased visibility or increased risk of predation (Davies et al., 2012; Gaston et al., 2013; Sullivan et al., 2019). Additionally, ALAN alters the structure of communities through both direct and indirect processes, such as modifications to biotic interactions like intraguild predation and adjustments to resource flows between adjacent ecosystems (Gal et al., 1999; Moore et al., 2000; Kuijper et al. 2008; Dwyer et al., 2013). Moreover, ALAN may cause changes in the feeding habits of invertebrate communities, which could alter the systems within communities (Manfrin et al., 2018; Sullivan et al., 2019). Finally, ALAN initially had a negative effect on the density of key riparian predators such as orb-web spiders. This negative impact on predators can lead to cascading effects on entire riparian ecosystems, potentially altering community composition (Sullivan et al., 2019).

Overall, the results suggest that invasive crayfish and the combined stressor had the most substantial impacts on altering the EAI's community composition over time, while the effects of ALAN alone were less pronounced. Further research is needed to elucidate the mechanisms driving these changes and their implications for ecosystem function and resilience.

4.4 Constraints and Future directions

While this study provides valuable insight into the combined impact of CR and ALAN on the EAI's community, it is important to acknowledge the limitations of the experimental approach. The accurate representation of natural macroinvertebrate communities is a major concern in mesocosm experiments. Research has indicated that the vulnerability and sensitivity of communities formed in mesocosms may differ from those in natural ecosystems. This is because of differences in habitat characteristics, environmental stress levels, and species establishment methods, which may affect the representativeness of the results (Reiber et al., 2022). The ecological dynamics and adaptive responses of EAI's communities to environmental change cannot be fully captured by mesocosm experiments because they are often short-term. The full range of stressors effects was not fully captured during the six-week study period. Over extended periods, antagonistic or synergistic interactions may gradually develop (Stewart et al.,

2013). The study focused on CR and ALAN as stressors but did not account for other potential environmental variables that may affect EAI communities. Fluctuations in water quality parameters (e.g., temperature, dissolved oxygen, and conductivity) can interact with experimental stressors, potentially masking or amplifying their effects (Bier et al., 2022).

Future research should aim to build on the findings of this study and address the remaining gaps in understanding of the interactive effects of multiple stressors on aquatic insect communities. Conducting field-based studies in natural watersheds would provide a more comprehensive assessment of how CR and ALAN affect EAI communities in their local habitats, accounting for the influence of broader environmental factors. This method will enhance predictive power and help to fill existing knowledge gaps (Reiber et al., 2022). Extending the temporal scope of the experiment, potentially over multiple seasons or years, can reveal long-term paths of community change and potential for recovery or adaptation (Stewart et al., 2013). By addressing these lines of research, future studies may provide more comprehensive insights to guide effective management strategies and conservation efforts for aquatic-terrestrial connectivity facing the challenges of invasive species and light pollution.

5. Conclusion

This study examined the combined effects of two common anthropogenic stressors, signal crayfish and ALAN, on EAIs in a stream mesocosm experiment. The results highlight the complex interplay of these stressors and their impact on EAIs abundance, biomass, and community structure over time.

Although individual stressors and their combinations did not have a significant impact on EAIs abundance and biomass among treatments, the study showed significant temporal dynamics. EAIs abundance decreased over time in the presence of crayfish, ALAN, and their combination, suggesting long-term deleterious effects of these stressors. Notably, combined stressor treatments showed greater decreases in frequency than individual stressors, suggesting a possible synergistic interaction.

In terms of community composition, the presence of signal crayfish and combined treatment of stressors resulted in the most significant deviations from control conditions, particularly after long-term exposure. These changes in community structure can be attributed to factors such as prey selection by crayfish, habitat changes, altered trophic dynamics, and competition for resources. ALAN alone had a less pronounced impact on community composition, possibly due to the moderating effect of the vegetation buffer.

The changes observed in EAIs communities under these stressors have important implications for ecosystem function and the maintenance of aquatic-terrestrial connectivity. Changes in EAIs abundance, biomass, and community composition can alter essential processes such as nutrient cycling, energy transfer, and food web dynamics between meta-ecosystems. This study highlights the importance of considering the combined effects of multiple stressors in environmental assessment and management strategies.

Although this study provides valuable information, future studies should consider limitations such as the representativeness of mesocosm experiments, the effects of additional environmental variables, and long-term trajectories of community change. Field studies in natural watersheds and over longer time periods will advance our understanding of the interactive effects of multiple stressors on aquatic insect communities and their impact on ecosystem resilience.

In summary, this study contributes to the growing body of knowledge about the impacts of multiple anthropogenic stressors on meta-ecosystems. Highlighting the combined impacts of invasive species and light pollution on EAIs communities need for comprehensive management approaches that consider the complex interplay between stressors and their potential to alter key ecosystem functions and connectivity.

6. References

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7. Appendix

7.1 Supplementary tables

Table S1. Overview of aquatic insects collected with emergence traps

Taxa	Abundance	Percentage	Taxa	Biomass (mg)	Percentage(mg)
Chironomidae	60770	97,839386914	Chironomidae	12478,79	93,67377946
Tanypodinae	440	0,708397733	Tanypodinae	350,9	2,63407984
Brachycera	192	0,309119011	Tipulidae	141,96	1,06564256
Simuliidae	179	0,288189078	Simuliidae	92,75	0,69624083
Dolichopodidae	100	0,160999485	Brachycera	75,76	0,56870302
Dixidae	93	0,149729521	Dixidae	40,89	0,30694649
Psychodidae	71	0,114309634	Muscidae	31,06	0,23315623
Ephydriidae	65	0,104649665	Baetidae	28,93	0,21716708
Empididae	40	0,064399794	Dolichopodidae	24,85	0,18653999
Baetidae	36	0,057959815	Ephydriidae	16,6	0,12461022
Ceratopogonidae	27	0,043469861	Scatohagidae	13,28	0,09968817
Muscidae	27	0,043469861	Psychodidae	5,08	0,03813373
Tipulidae	24	0,038639876	Empididae	4,53	0,03400508
Scatohagidae	15	0,024149923	Sciomyzidae	2,83	0,02124379
Sciomyzidae	8	0,012879959	Ephemereptera	2,68	0,02011779
Culicidae	6	0,009659969	Diptera	2,53	0,0189918
Diptera	6	0,009659969	Limoniidae	2,33	0,01749047
Limoniidae	6	0,009659969	Culicidae	2,27	0,01704007
Ephemereptera	2	0,00321999	Ceratopogonidae	1,91	0,01433768
Trichoptera	2	0,00321999	Plecoptera	1,61	0,01208569
Heptageniidae	1	0,001609995	Heptageniidae	0	0
Hydroppsiidae	1	0,001609995	Hydroppsiidae	0	0
Plecoptera	1	0,001609995	Trichoptera	0	0

Table S2. Linear mixed-effects models (LMMs) obtained with stressors abundance contrasts values, estimate, standard Error (SE), degrees of Freedom (df), lower Confidence Limit (lower.CL), upper Confidence Limit (upper.CL), t.ratio, p.value,

contrast	Treatment	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
Week_0 - Week_1	CO	-1.49346777	0.3350109	36	-2.3955400	-0.59139559	-4.4579672	0.0004649022
Week_0 - Week_1	CR	-1.71565580	0.3350109	36	-2.6175323	-0.81377925	-5.1211934	0.00006710076
Week_0 - Week_1	ALAN	-1.52662209	0.3350109	36	-2.4294710	-0.62377317	-4.5569321	0.0003150914
Week_0 - Week_1	ALAN+CR	-2.11546280	0.3350109	36	-3.0182411	-1.21268450	-6.3146082	0.00000517187
Week_0 - Week_4	CO	-0.31392849	0.3350109	36	-1.2160007	0.58814369	-0.9370694	0.785225
Week_0 - Week_4	CR	-0.07195025	0.3350109	36	-0.9738268	0.82992629	-0.2147699	0.9964358
Week_0 - Week_4	ALAN	-0.16333123	0.3350109	36	-1.0661802	0.73951769	-0.4875400	0.9613831
Week_0 - Week_4	ALAN+CR	-0.95624225	0.3350109	36	-1.8590206	-0.05346395	-2.8543613	0.03438844
Week_0 - Week_6	CO	-0.93481891	0.3350109	36	-1.8368911	-0.03274673	-2.7904131	0.03994568
Week_0 - Week_6	CR	-1.42959162	0.3350109	36	-2.3314682	-0.52771508	-4.2672984	0.0007341948
Week_0 - Week_6	ALAN	-0.44489414	0.3350109	36	-1.3477431	0.45795479	-1.3279988	0.5515712

Week_0 - Week_6	ALAN+CR	-2.00318487	0.3350109	36	-2.9059632	-1.10040656	-5.9794611	0.000002652274
Week_1 - Week_4	CO	1.17953928	0.3350109	36	0.2774671	2.08161147	3.5208978	0.006179889
Week_1 - Week_4	CR	1.64370555	0.3350109	36	0.7418290	2.54558209	4.9064235	0.0000968246
Week_1 - Week_4	ALAN	1.36329086	0.3350109	36	0.4604419	2.26613979	4.0693921	0.001319343
Week_1 - Week_4	ALAN+CR	1.15922055	0.3350109	36	0.2564422	2.06199885	3.4602469	0.007529341
Week_1 - Week_6	CO	0.55864886	0.3350109	36	-0.3434233	1.46072104	1.6675541	0.3553021
Week_1 - Week_6	CR	0.28606417	0.3350109	36	-0.6158124	1.18794071	0.8538950	0.8282567
Week_1 - Week_6	ALAN	1.08172796	0.3350109	36	0.1788790	1.98457688	3.2289333	0.01383052
Week_1 - Week_6	ALAN+CR	0.11227793	0.3350109	36	-0.7905004	1.01505624	0.3351471	0.986817
Week_4 - Week_6	CO	-0.62089043	0.3350109	36	-1.5229626	0.28118176	-1.8533438	0.2659028
Week_4 - Week_6	CR	-1.35764137	0.3350109	36	-2.2595179	-0.45576483	-4.0525285	0.001344095
Week_4 - Week_6	ALAN	-0.28156291	0.3350109	36	-1.1844118	0.62128601	-0.8404589	0.8348112
Week_4 - Week_6	ALAN+CR	-1.04694261	0.3350109	36	-1.9497209	-0.14416431	-3.1250998	0.01741353

Table S3: PERMANOVA Comparisons for Treatment, Week, Community Composition; sums of squares (SS), R-squared, F value, significant p-values

pairs	Df	SS	F,Model	R ²	p,value
ALAN,Week_0 vs ALAN,Week_1	1	0,1509553687	12,53307250	0,676254436	0,022
ALAN,Week_0 vs ALAN,Week_4	1	0,0895682879	4,49509814	0,428304536	0,034
ALAN,Week_0 vs ALAN,Week_6	1	0,0395313676	1,96183738	0,246405105	0,203
ALAN,Week_0 vs ALAN+CR,Week_0	1	0,1131428506	4,69600025	0,439042646	0,029
ALAN,Week_0 vs ALAN+CR,Week_1	1	0,0905043696	6,50128957	0,520049515	0,038
ALAN,Week_0 vs ALAN+CR,Week_4	1	0,0626972405	2,07816578	0,257257135	0,190
ALAN,Week_0 vs ALAN+CR,Week_6	1	0,0911243744	4,59356913	0,433618649	0,058
ALAN,Week_0 vs CO,Week_0	1	0,0134577569	0,63697060	0,095973093	0,761
ALAN,Week_0 vs CO,Week_1	1	0,1638244768	14,66395308	0,709639294	0,029
ALAN,Week_0 vs CO,Week_4	1	0,1067759047	4,62336095	0,435206991	0,057
ALAN,Week_0 vs CO,Week_6	1	0,0838145858	4,78041358	0,443435082	0,082
ALAN,Week_0 vs CR,Week_0	1	0,0264148881	1,50753601	0,200803034	0,336
ALAN,Week_0 vs CR,Week_1	1	0,0958761026	7,80158704	0,565267387	0,030
ALAN,Week_0 vs CR,Week_4	1	0,0611238377	2,22161232	0,270216136	0,116
ALAN,Week_0 vs CR,Week_6	1	0,0935753382	7,02869261	0,534977968	0,038
ALAN,Week_1 vs ALAN,Week_4	1	0,1439531821	9,39927178	0,610371186	0,025
ALAN,Week_1 vs ALAN,Week_6	1	0,0756339972	4,86712738	0,447876169	0,024
ALAN,Week_1 vs ALAN+CR,Week_0	1	0,3621993047	18,59049578	0,756003293	0,038
ALAN,Week_1 vs ALAN+CR,Week_1	1	0,0182104424	1,95588849	0,245841617	0,113
ALAN,Week_1 vs ALAN+CR,Week_4	1	0,1361846274	5,32822588	0,470349544	0,031
ALAN,Week_1 vs ALAN+CR,Week_6	1	0,0369860936	2,42898654	0,288170652	0,068
ALAN,Week_1 vs CO,Week_0	1	0,1034485066	6,26302549	0,510724331	0,042
ALAN,Week_1 vs CO,Week_1	1	0,0030102063	0,45876778	0,071030233	0,892
ALAN,Week_1 vs CO,Week_4	1	0,1068245898	5,77915906	0,490625776	0,022
ALAN,Week_1 vs CO,Week_6	1	0,0284160841	2,19896168	0,268200019	0,033
ALAN,Week_1 vs CR,Week_0	1	0,2241398149	17,35972942	0,743147710	0,029
ALAN,Week_1 vs CR,Week_1	1	0,0068771396	0,89558995	0,129878656	0,574
ALAN,Week_1 vs CR,Week_4	1	0,1941501757	8,47711348	0,585552741	0,026
ALAN,Week_1 vs CR,Week_6	1	0,0459063203	5,27481787	0,467840628	0,028
ALAN,Week_4 vs ALAN,Week_6	1	0,0351731066	1,50177849	0,200189661	0,258
ALAN,Week_4 vs ALAN+CR,Week_0	1	0,2548661747	9,31383960	0,608197542	0,022
ALAN,Week_4 vs ALAN+CR,Week_1	1	0,1066246192	6,20206975	0,508280142	0,028
ALAN,Week_4 vs ALAN+CR,Week_4	1	0,0252873180	0,75619288	0,111925887	0,541
ALAN,Week_4 vs ALAN+CR,Week_6	1	0,0715165623	3,09486017	0,340286724	0,050
ALAN,Week_4 vs CO,Week_0	1	0,0911210377	3,73469149	0,383647647	0,027
ALAN,Week_4 vs CO,Week_1	1	0,1591761942	11,02121339	0,647498691	0,031
ALAN,Week_4 vs CO,Week_4	1	0,0107941067	0,40940022	0,063874965	0,774
ALAN,Week_4 vs CO,Week_6	1	0,0691819288	3,32546145	0,356600203	0,070
ALAN,Week_4 vs CR,Week_0	1	0,1414764089	6,80414232	0,531401647	0,025
ALAN,Week_4 vs CR,Week_1	1	0,1143592702	7,34952009	0,550545640	0,036
ALAN,Week_4 vs CR,Week_4	1	0,0889983812	2,89105301	0,325164298	0,116

ALAN,Week_4 vs CR,Week_6	1	0,0673864059	4,06330724	0,403774539	0,026
ALAN,Week_6 vs ALAN+CR,Week_0	1	0,2254036570	8,17015902	0,576574971	0,027
ALAN,Week_6 vs ALAN+CR,Week_1	1	0,0393800313	2,26111677	0,273705944	0,051
ALAN,Week_6 vs ALAN+CR,Week_4	1	0,0279690459	0,83081208	0,121627131	0,564
ALAN,Week_6 vs ALAN+CR,Week_6	1	0,0190063165	0,81458276	0,119535237	0,406
ALAN,Week_6 vs CO,Week_0	1	0,0454210338	1,84466246	0,235148736	0,162
ALAN,Week_6 vs CO,Week_1	1	0,0837957306	5,71316989	0,487756085	0,036
ALAN,Week_6 vs CO,Week_4	1	0,0389589659	1,46516997	0,196267462	0,271
ALAN,Week_6 vs CO,Week_6	1	0,0135151605	0,64271861	0,096755357	0,479
ALAN,Week_6 vs CR,Week_0	1	0,0991851681	4,71926165	0,440259955	0,063
ALAN,Week_6 vs CR,Week_1	1	0,0538018606	3,40852339	0,362280376	0,057
ALAN,Week_6 vs CR,Week_4	1	0,0577022329	1,86085347	0,236724101	0,187
ALAN,Week_6 vs CR,Week_6	1	0,0133324254	0,79319382	0,116763020	0,355
ALAN+CR,Week_0 vs ALAN+CR,Week_1	1	0,2936889294	13,74982827	0,696199890	0,035
ALAN+CR,Week_0 vs ALAN+CR,Week_4	1	0,1515161698	4,02882999	0,401724827	0,054
ALAN+CR,Week_0 vs ALAN+CR,Week_6	1	0,3099569969	11,36378711	0,654453262	0,032
ALAN+CR,Week_0 vs CO,Week_0	1	0,1432784738	5,01565982	0,455320871	0,031
ALAN+CR,Week_0 vs CO,Week_1	1	0,3807502714	20,45901403	0,773234180	0,035
ALAN+CR,Week_0 vs CO,Week_4	1	0,3003484154	9,83673603	0,621134053	0,028
ALAN+CR,Week_0 vs CO,Week_6	1	0,3003139297	12,02632035	0,667153369	0,034
ALAN+CR,Week_0 vs CR,Week_0	1	0,0444023059	1,77891219	0,228683928	0,116
ALAN+CR,Week_0 vs CR,Week_1	1	0,2829155348	14,34097033	0,705028821	0,021
ALAN+CR,Week_0 vs CR,Week_4	1	0,1076125600	3,07888882	0,339126173	0,046
ALAN+CR,Week_0 vs CR,Week_6	1	0,3225690873	15,54414291	0,721502033	0,032
ALAN+CR,Week_1 vs ALAN+CR,Week_4	1	0,0870460260	3,17274932	0,345888589	0,029
ALAN+CR,Week_1 vs ALAN+CR,Week_6	1	0,0190633958	1,11459747	0,156663462	0,326
ALAN+CR,Week_1 vs CO,Week_0	1	0,0570104054	3,09944222	0,340618924	0,061
ALAN+CR,Week_1 vs CO,Week_1	1	0,0242763283	2,87704831	0,324099657	0,020
ALAN+CR,Week_1 vs CO,Week_4	1	0,0685853583	3,36848734	0,359555093	0,020
ALAN+CR,Week_1 vs CO,Week_6	1	0,0242050038	1,63559191	0,214206302	0,260
ALAN+CR,Week_1 vs CR,Week_0	1	0,1575391362	10,65324151	0,639709783	0,032
ALAN+CR,Week_1 vs CR,Week_1	1	0,0067334194	0,70467765	0,105102391	0,637
ALAN+CR,Week_1 vs CR,Week_4	1	0,1259813766	5,08414006	0,458686017	0,023
ALAN+CR,Week_1 vs CR,Week_6	1	0,0288399044	2,72605746	0,312404253	0,048
ALAN+CR,Week_4 vs ALAN+CR,Week_6	1	0,0641745847	1,92416510	0,242822439	0,179
ALAN+CR,Week_4 vs CO,Week_0	1	0,0580776822	1,67649690	0,218393484	0,234
ALAN+CR,Week_4 vs CO,Week_1	1	0,1465367245	5,93591727	0,497315551	0,022
ALAN+CR,Week_4 vs CO,Week_4	1	0,0322680148	0,88141340	0,128086099	0,323
ALAN+CR,Week_4 vs CO,Week_6	1	0,0698816261	2,25080101	0,272797878	0,184
ALAN+CR,Week_4 vs CR,Week_0	1	0,0785011799	2,52932396	0,296544483	0,190
ALAN+CR,Week_4 vs CR,Week_1	1	0,0946648581	3,66863454	0,379436675	0,034
ALAN+CR,Week_4 vs CR,Week_4	1	0,0298024704	0,72639680	0,107991964	0,527
ALAN+CR,Week_4 vs CR,Week_6	1	0,0635005693	2,36696302	0,282893926	0,171
ALAN+CR,Week_6 vs CO,Week_0	1	0,0739882108	3,04351042	0,336540821	0,070
ALAN+CR,Week_6 vs CO,Week_1	1	0,0394224556	2,74638250	0,314002103	0,031
ALAN+CR,Week_6 vs CO,Week_4	1	0,0513188496	1,95297494	0,245565333	0,149
ALAN+CR,Week_6 vs CO,Week_6	1	0,0087342952	0,42163461	0,065658456	0,717
ALAN+CR,Week_6 vs CR,Week_0	1	0,1737684076	8,39286623	0,583126814	0,033
ALAN+CR,Week_6 vs CR,Week_1	1	0,0252137984	1,62967101	0,213596498	0,172
ALAN+CR,Week_6 vs CR,Week_4	1	0,1204158764	3,92289309	0,395337635	0,067
ALAN+CR,Week_6 vs CR,Week_6	1	0,0008814189	0,05343313	0,008826913	1,000
CO,Week_0 vs CO,Week_1	1	0,1142884165	7,30525131	0,549050231	0,028
CO,Week_0 vs CO,Week_4	1	0,0995488747	3,61107684	0,375720317	0,079
CO,Week_0 vs CO,Week_6	1	0,0831419845	3,77820379	0,386390371	0,079
CO,Week_0 vs CR,Week_0	1	0,0442234798	2,01064521	0,250996662	0,114
CO,Week_0 vs CR,Week_1	1	0,0562293842	3,35455823	0,358601459	0,040
CO,Week_0 vs CR,Week_4	1	0,0787722711	2,46270676	0,291006983	0,093
CO,Week_0 vs CR,Week_6	1	0,0822678724	4,62539983	0,435315367	0,027
CO,Week_1 vs CO,Week_4	1	0,1179197108	6,69549151	0,527391279	0,033
CO,Week_1 vs CO,Week_6	1	0,0341308020	2,83246544	0,320687973	0,031
CO,Week_1 vs CR,Week_0	1	0,2391040160	19,86105669	0,767990919	0,040
CO,Week_1 vs CR,Week_1	1	0,0131072740	1,92577060	0,242975819	0,165
CO,Week_1 vs CR,Week_4	1	0,1990283034	9,03433137	0,600913413	0,017
CO,Week_1 vs CR,Week_6	1	0,0499392602	6,37771380	0,515257818	0,021
CO,Week_4 vs CO,Week_6	1	0,0594921109	2,48164993	0,292590469	0,067
CO,Week_4 vs CR,Week_0	1	0,1778756042	7,42330340	0,553016137	0,047
CO,Week_4 vs CR,Week_1	1	0,0789609166	4,21592624	0,412681742	0,024
CO,Week_4 vs CR,Week_4	1	0,0880649543	2,59371829	0,301815606	0,098
CO,Week_4 vs CR,Week_6	1	0,0482375693	2,44200980	0,289268771	0,077

CO,Week_6 vs CR,Week_0	1	0,1654889078	8,99404348	0,599841096	0,028
CO,Week_6 vs CR,Week_1	1	0,0280265756	2,12850697	0,261857064	0,109
CO,Week_6 vs CR,Week_4	1	0,1219618877	4,29576117	0,417235899	0,076
CO,Week_6 vs CR,Week_6	1	0,0125473550	0,88415989	0,128433957	0,418
CR,Week_0 vs CR,Week_1	1	0,1576231727	11,98088016	0,666312219	0,035
CR,Week_0 vs CR,Week_4	1	0,0559007398	1,96970935	0,247149459	0,187
CR,Week_0 vs CR,Week_6	1	0,1772885297	12,50249347	0,675719383	0,036
CR,Week_1 vs CR,Week_4	1	0,1471130949	6,35543361	0,514383696	0,035
CR,Week_1 vs CR,Week_6	1	0,0367441062	4,10655763	0,406326049	0,023
CR,Week_4 vs CR,Week_6	1	0,1197887966	4,95575861	0,452342808	0,033

7.2 Supplementary figures

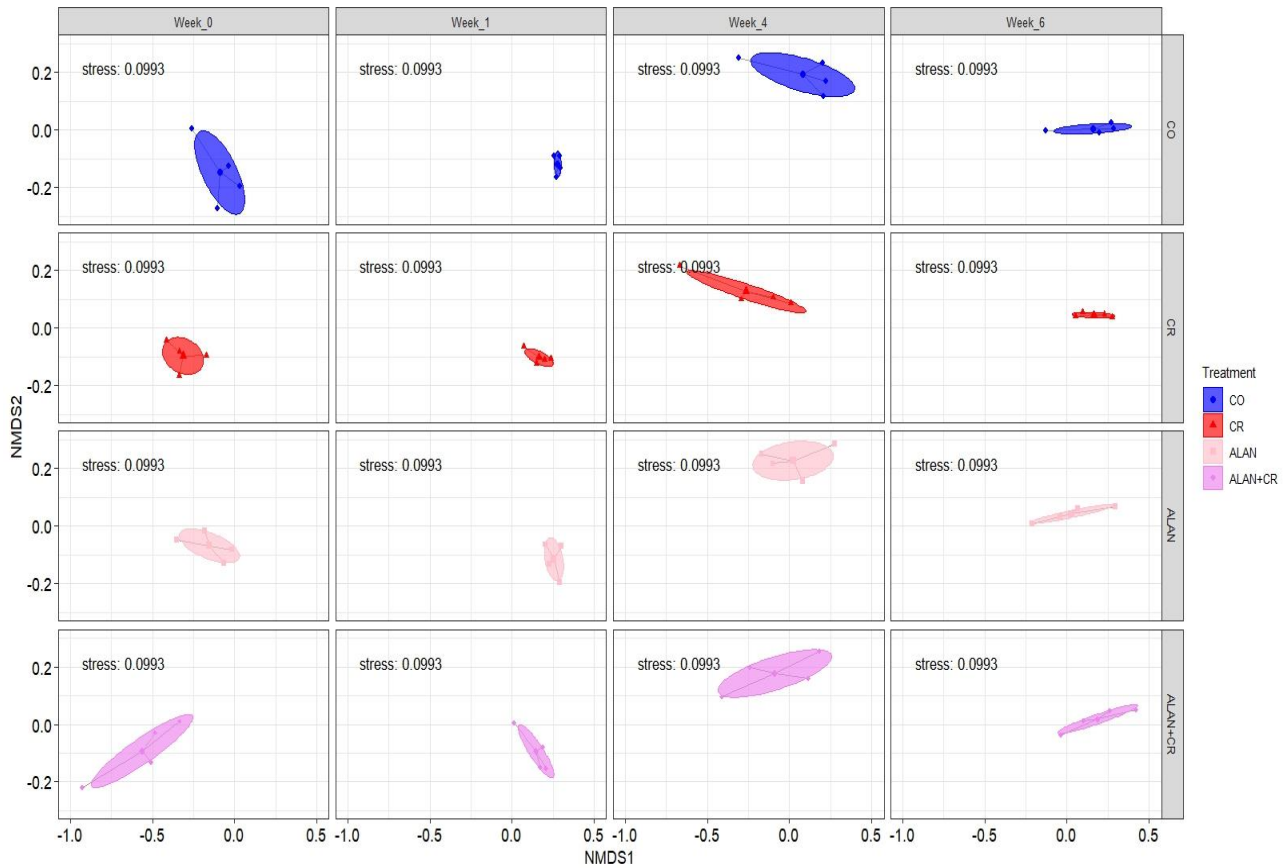


Figure S1: Facetted NMDS plot depicting the temporal dynamics of emergency aquatic insect abundance communities under different treatments. Each facet represents a specific treatment (CO, ALAN, CR, ALAN + CR), and each point represents a sample. The arrangement of points within each facet illustrates treatment-specific responses over the six-week study period.